

Machine learning research project

(15% of final grade. Deadline: 5pm 22 April 2023)

Goal

The goal of this project is to apply various machine learning models with real data and consider a variety of model specifications and obtain a predictive model as accurate as possible for the classification of asset return trends. This means you will explore a variety of machine learning techniques and features to design your models. You are required to compare the following machine learning algorithms as a result of your implementation, these are: SVM, logistic regression, and artificial neural networks that we have discussed in lectures with MATLAB examples for classification problems. For this purpose, you can use all the available MATLAB examples and files on LMO.

Literature

In order to write your report properly you will also need to search for more articles other than the ones on LMO and include these in your introduction or in a separate literature review section. You do not need to refer to all the previous work on machine learning, however, you can include some earlier work that use the above algorithms in the classification of stock or futures prices.

Design Matrix and the Target

To design your input matrix X, feel free to work with the past lagged values of prices or **log-returns**, technical indicators such as **moving averages**, **relative strength index**, **stochastic oscillator**, etc. These indicators are available on MATLAB. Furthermore, you can have your target variable defined as the prediction of up or down trend in the next 5-minutes or longer time horizons such as 20-minutes.

Dataset

You will use the 5-minute data of most active soybean futures contracts as given in “**a5min.csv**” file on LMO. Alternatively, you may use trading data on Chinese market.

Figures, Tables and Code

Figures and Tables of your results should be within your manuscript. **All your MATLAB codes should be added as the appendix of your report.**

Possible Sections in the Report

The research report should follow typical sections as in many scientific articles (see some sample articles on LMO) such as introduction, methodology, dataset, empirical results, conclusion and reference sections. However, feel free to change these as needed. There is no requirement on the length of your report.

Marking Criteria

The marking criteria is based on four aspects of your work:

- Understanding and using appropriate machine learning models (25 points)
- Presentation of results clearly and demonstration of understanding (30 points)
- Creativity in the design of the input matrix and creation of new features (25 points)
- Overall quality of written report with appropriate references (20 points)